DATA ANALYTICS EXCEL

PROJECT REPORT

Practical Assignments: Superstore Sales, Adecco HR Analytics, and Porter Delivery

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A professional, business-focused summary of the analysis, findings, and recommendations

1. Executive Summary

This report brings together three practical data analytics assignments completed using Excel-based analysis: (1) Superstore Sales, (2) Adecco HR Analytics, and (3) Porter Delivery Analytics. Each project started with a supplied dataset, followed by data review, calculations, summary tables, and interpretation of the results. The purpose was not only to calculate metrics, but also to explain what the findings could mean for business decisions.

Assignment	Dataset scope	Headline finding
Superstore Sales	9,994 order-line records; 21 columns	Total sales were approximately $2.30 million and total profit was approximately $286,397.
Adecco HR Analytics	1,470 employee records	237 employees left, giving an overall attrition rate of 16.12%; Sales had the highest departmental rate at 20.63%.
Porter Delivery Analytics	197,428 delivery records; 14 source columns	Average delivery time was about 48.47 minutes; only 14.15% of valid deliveries were completed within 30 minutes.

The projects demonstrate progression from spreadsheet fundamentals and business reporting to workforce analysis and operational analytics. The findings should be interpreted in context: unusual values can affect averages, correlation does not prove causation, and some questions require fields that are not present in the supplied data.

2. Approach and Tools

The analysis was carried out using Excel techniques suited to each assignment. These included basic formulas, date/time calculations, IF and conditional aggregation functions, lookup methods, sorting and filtering, PivotTables, summary statistics, correlation, outlier checks, and regression where required. The original source data was retained for traceability, while helper columns and summary tables were used to make the analysis easier to review.

- Reviewed the dataset structure and checked key fields before calculating results.
- Used formulas and PivotTables to summarize records by category, region, department, market, day, or another relevant group.
- Used averages, rates, correlations, variance/standard deviation, and outlier thresholds only were relevant to the question.
- Translated the numerical results into plain-language observations and practical recommendations.
- Recorded data limitations instead of inventing missing fields or treating association as proof of cause.

3. Assignment 1— Superstore Sales Analysis

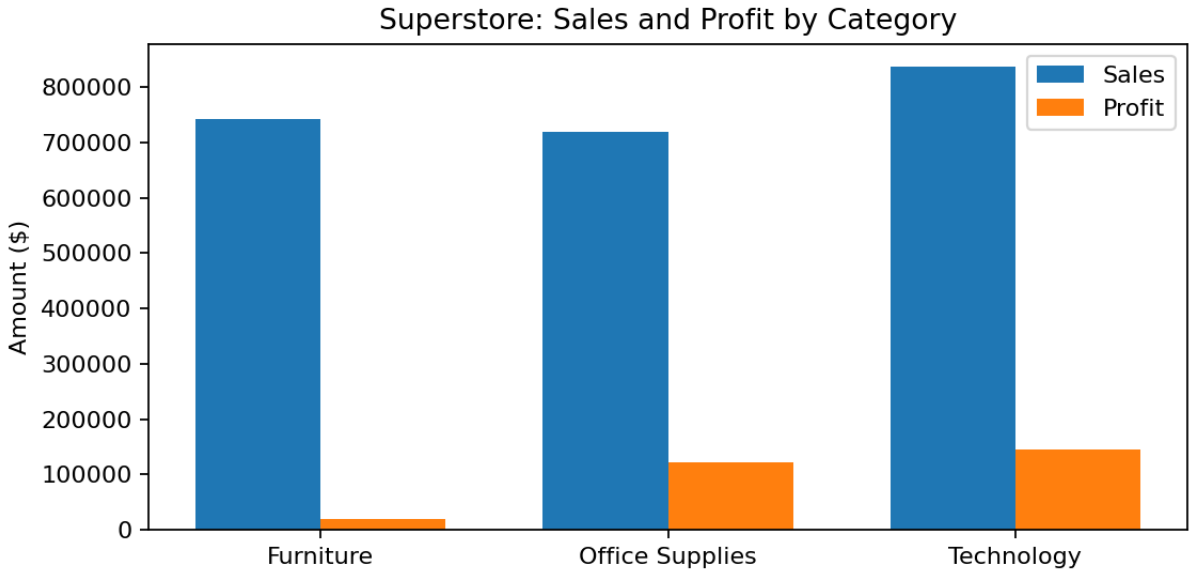
3.1 Objective

The objective was to complete an end-to-end Excel assignment using the Superstore sales dataset. The work covered spreadsheet formulas, data formatting, sorting and filtering, date functions, logical functions, data cleaning, PivotTables, lookup formulas, conditional aggregation, and sales/profit summaries.

3.2 Dataset and overall results

Metric	Result	Interpretation
Order rows	9,994	The dataset contains 9,994 order-line records.
Total sales	$2,297,200.86	Overall sales value across the supplied records.
Total profit	$286,397.02	Overall recorded profit across the supplied records.
Total quantity	37,873	Total units recorded.
Unique customers	793	Distinct customer counts in the dataset.
Unique orders	5,009	Distinct order count in the dataset.
Unique products	1,862	Distinct product counts in the dataset.

3.3 Category and regional findings



Technology generated the highest sales at approximately $836,154 and the highest profit at approximately $145,455. Office Supplies generated approximately $719,047 in sales and $122,491 in profit. Furniture generated approximately $742,000 in sales but only about $18,451 in profit. This difference suggests that sales volume alone does not tell the full story; profitability should also be reviewed when deciding where to focus commercial effort.

Category	Sales	Profit
Furniture	$741,999.80	$18,451.27
Office Supplies	$719,047.03	$122,490.80
Technology	$836,154.03	$145,454.95

By region, the West recorded the highest sales at approximately $725,458. The East recorded approximately $678,781 in sales and $91,523 in profit, while the West recorded approximately $108,418 in profit. These figures can support regional performance reviews, but a fuller business assessment would also consider discounting, product mix, shipping costs, and the number of orders.

3.4 Trend and Excel skills demonstrated

Annual sales were approximately $484,247 in 2014, $470,533 in 2015, $609,206 in 2016, and $733,215 in 2017. Sales dipped slightly in 2015 and then increased in 2016 and 2017. The calculated compound annual growth rate from 2014 to 2017 is approximately 14.8%. The assignment also practiced COUNT, COUNTA, COUNTBLANK, date functions, logical formulas, lookup formulas, conditional sums/averages/counts, conditional formatting, and PivotTable summaries.

3.5 Recommendations

- Review Furniture profitability separately from sales because its recorded profit is comparatively low.
- Use category, region, and time-based PivotTables as a repeatable monthly reporting process.
- Investigate whether discounts, product mix, or fulfilment costs explain differences in category-level profitability.
- Keep formulas and lookup tables documented so the workbook can be updated consistently.

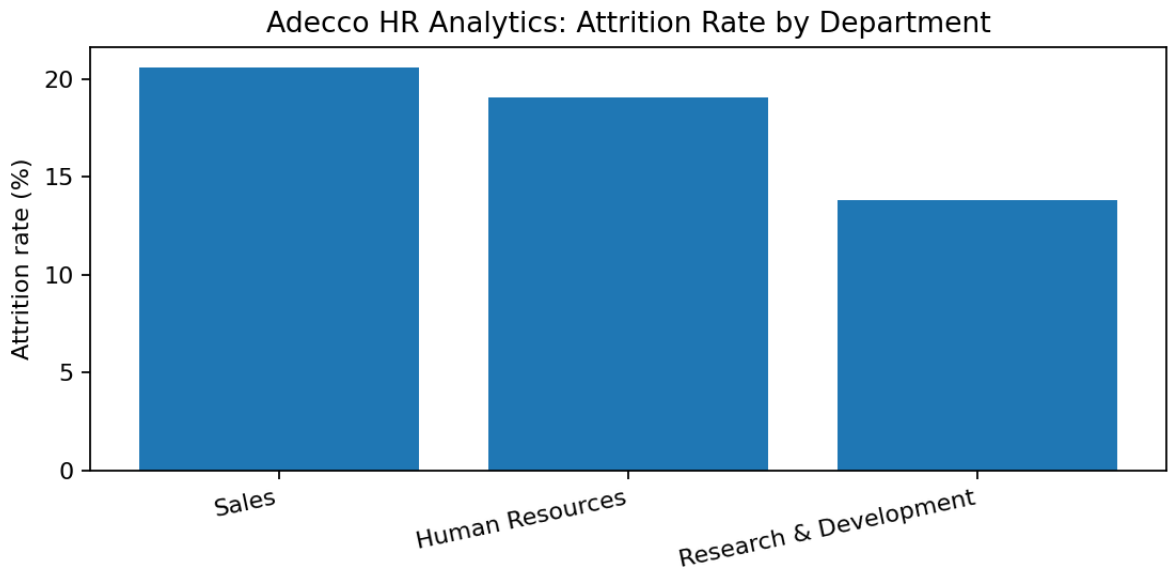
4. Assignment 2 — Adecco HR Analytics

4.1 Objective

This assignment examined employee attrition and the relationship between attrition and selected workforce characteristics. The analysis used counts, attrition rates, PivotTables, averages, and correlations to identify areas that may deserve further attention from HR.

4.2 Main findings

Metric	Result	Interpretation
Total employees	1,470	Total employee records in the dataset.
Employees who left	237	Employees with Attrition = Yes.
Overall attrition rate	16.12%	Approximately 16 out of every 100 employees left.
Average age of employees who left	33.6 years	Describes the average age within the attrition group.
Average monthly income of employees who left	4,787.09	Dataset currency/unit is not assumed beyond the supplied field.
Employees working overtime	416	Employees marked Overtime = Yes.
Average years at company	7.01 years	Average tenure across the employee dataset.



Sales had the highest departmental attrition rate at 20.63% (92 departures from 446 employees). Human Resources had a rate of 19.05% (12 departures among 63 employees), and Research & Development had a rate of 13.84% (133 departures among 961 employees). Although Research & Development had the largest number of departures, Sales had the highest rate. This distinction matters because department size affects the number of departures.

Department	Employees	Attrition = Yes	Attrition rate
Sales	446	92	20.63%
Human Resources	63	12	19.05%
Research & Development	961	133	13.84%

4.3 Additional observations

Male attrition was 17.01%, compared with 14.80% for female employees in the supplied dataset. The strongest absolute correlations among the tested numeric variables were Job Level (-0.169), Monthly Income (-0.160), and Age (-0.159). These are modest relationships, not evidence that any single factor causes an employee to leave. The correlation between DistanceFromHome and the attrition flag was approximately 0.078, which is weak.

4.4 Recommendations

- Begin a focused retention review in Sales, where the attrition rate is highest.
- Review overtime, job involvement, satisfaction, compensation, tenure, and career progression together rather than relying on one metric.
- Compare both counts and rates; the department with the largest number of departures is not always the department with the highest attrition rate.
- Treat correlations as signals for further investigation, not as causal conclusions, and handle employee-level analysis sensitively.

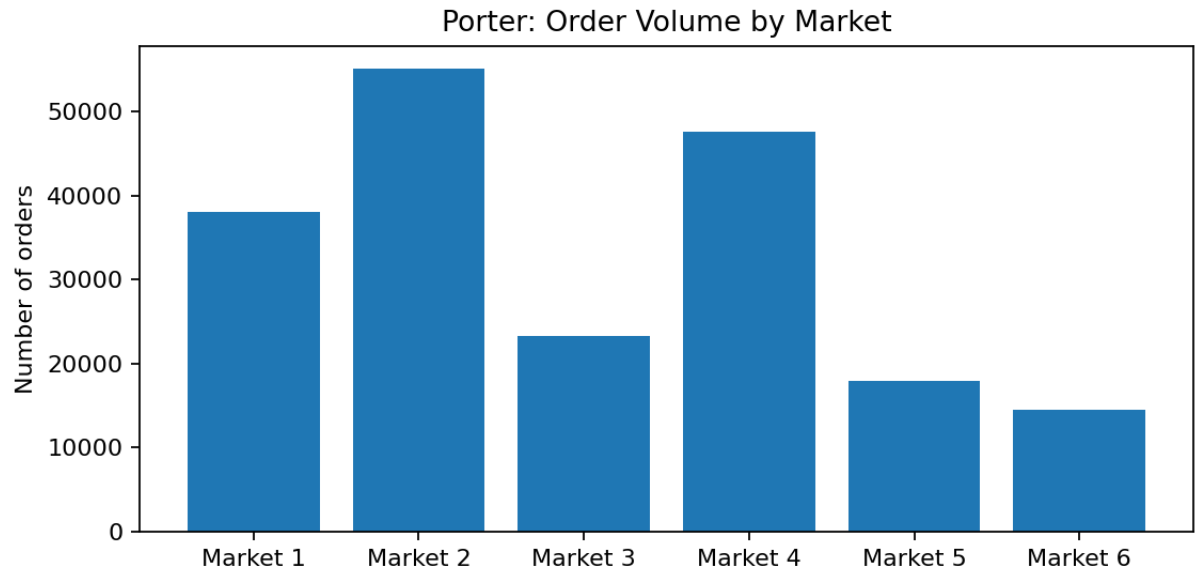
5. Assignment 3 — Porter Delivery Analytics

5.1 Objective

The Porter assignment focused on delivery duration, order demand, restaurant categories, market differences, partner availability, and unusual delivery records. The source dataset contains 197,428 rows and 14 columns. Delivery duration was derived by subtracting the order creation timestamp from the actual delivery timestamp and converting the difference to minutes.

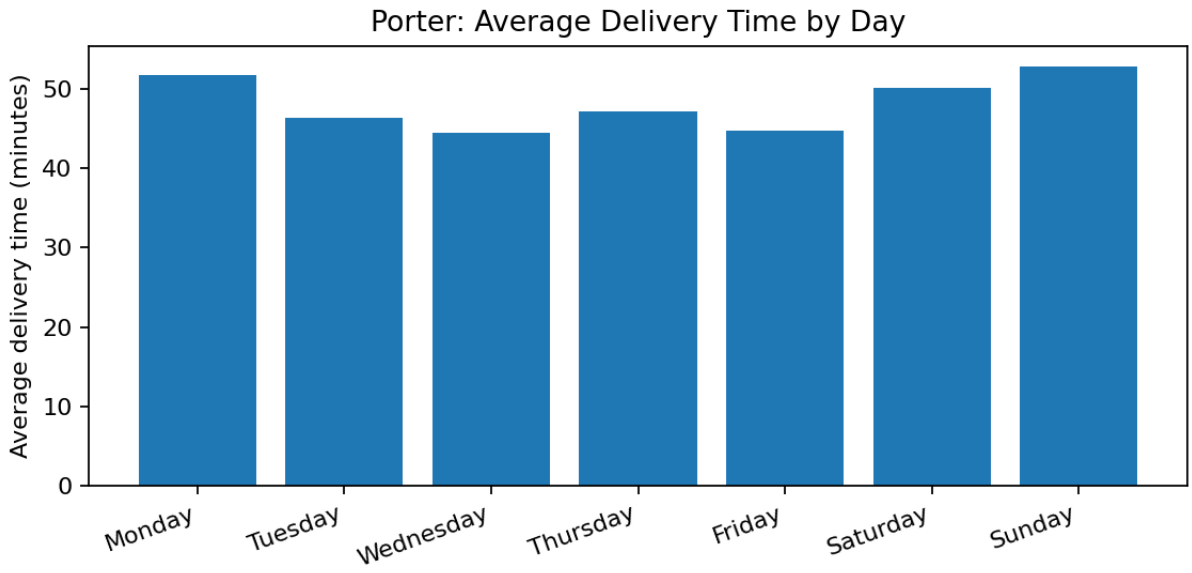
5.2 Overall delivery performance

Metric	Result	Interpretation
Total records	197,428	Number of rows in the source dataset.
Valid delivery-time records	197,421	Seven rows lacked an actual delivery timestamp.
Average delivery time	48.47 minutes	Mean duration across valid delivery records; sensitive to extreme values.
Median delivery time	44 minutes	Middle delivery duration and less sensitive to extreme values.
Deliveries completed within 30 minutes	27,938 (14.15%)	Only a minority of valid deliveries met a 30-minute target.
Upper IQR outlier fence	87.5 minutes	Delivery durations above this threshold were flagged as high outliers.
High delivery-time outliers	6,589	Records above the IQR upper fence; they should be investigated before operational conclusions.



Market 2 had the highest recorded order volume with 55,058 orders, followed by Market 4 with 47,599 and Market 1 with 38,037. The most common recorded order hour was 2 AM, with 36,976 orders, followed by 1 AM and 3 AM. Saturday had the highest daily order volume at 34,541 orders.

5.3 Delivery time by day and category



Sunday had the highest average delivery time at approximately 52.72 minutes, followed by Monday at 51.72 minutes and Saturday at 50.09 minutes. Wednesday had the lowest average at about 44.47 minutes. Italian was the highest-volume restaurant category in the category analysis and had an average delivery time of approximately 70.20 minutes. This figure should be interpreted carefully because extreme delivery times can pull averages upward.

5.4 Order value, partners, and statistical analysis

The correlation between total items and subtotal was approximately 0.558, indicating a moderate positive relationship: orders with more items generally tended to have higher subtotals. The correlation between item price range and subtotal was approximately 0.510, also a moderate positive relationship. Market 4 had the highest average subtotal at approximately 2,896.37.

Partner availability was assessed using average busy partners, average on-shift partners, and their ratio. Market 1 had a busy-to-the-on-shift ratio of approximately 0.971 and an average delivery time of 55.21 minutes. This pattern suggests that partner workload may be worth investigating, but the relationship does not establish causation. The cleaned regression using total items and num_distinct_items had an R-squared of approximately 0.0286, meaning these two variables explained only a small share of delivery-time variation in that model.

5.5 Data quality and limitations

The source data includes missing values in market_id (987), store_primary_category (4,760), order protocol (995), and the three partner-related fields (16,262 each). Seven records have no actual delivery timestamp and cannot be used to calculate delivery duration. Extreme delivery-time records substantially affect some averages and standard deviations, so raw and outlier-aware results should be distinguished.

The assignment asks for first-time versus repeat-customer analysis, but the supplied dataset does not include customer_id. That question cannot be answered reliably from this file alone. A customer identifier or an order history table containing customer identifiers would be needed; no customer IDs were fabricated.

5.6 Recommendations

- Investigate high-delivery-time outliers and confirm whether they reflect genuine delays, timestamp issues, or unusual cases.
- Review staffing and partner availability around high-volume hours and weekend periods.
- Compare category averages alongside order counts and outlier-adjusted summaries to avoid overreacting to small samples or extreme values.
- Add customer_id to the data model if first-time versus repeat-customer analysis is required.
- Track the 30-minute delivery rate as an operational KPI and monitor it by market, day, hour, and category.

6. Cross-Project Skills and Learning

Across the three assignments, the work demonstrates practical skills in spreadsheet-based data analysis and business communication:

- Data preparation and quality checks, including missing values, timestamps, outliers, and helper columns.
- Excel formulas such as AVERAGE, COUNT, IF, IFERROR, CORREL, quartiles, and conditional aggregation.
- PivotTables for comparing categories, regions, departments, markets, and time periods.
- Descriptive statistics include averages, medians, rates, variance, standard deviation, and correlations.
- Turning numerical results into clear findings, operational recommendations, and documented limitations.

7. Overall Conclusion

These three assignments show how data analysis can support different business functions. The Superstore project focused on sales and profitability, the Adecco project focused on employee attrition, and the Porter project focused on delivery performance and operational capacity. In each case, the strongest outcome is not simply the calculation itself, but the ability to explain what the number means, what it does not prove, and what should be investigated next.

The next step for each project would be to refresh the analysis with updated data, validate unusual records with the relevant business team, and convert the most useful metrics into a repeatable dashboard or periodic report.